

Financial Distress Warning — What "Getting the Data Right" Is Worth

An end-to-end reproducible LightGBM deployment: **public filings** → **a risk list**, with **who could see what, and when** turned into a measurable score gap · 2026-09 · local repo `C:\dev\finlab-ml` (33 commits · 10 commands to rerun · public repo pending)

Problem

- “Great offline, useless in production” is rarely a modelling problem — it is **misused data**: statements that were not published yet, training samples from the future, the *latest* filing date mistaken for the *first* one.
- Such mistakes **raise no alarm**: the pipeline runs and the score looks better ⇒ every cell needs a criterion that **needs no human referee**.

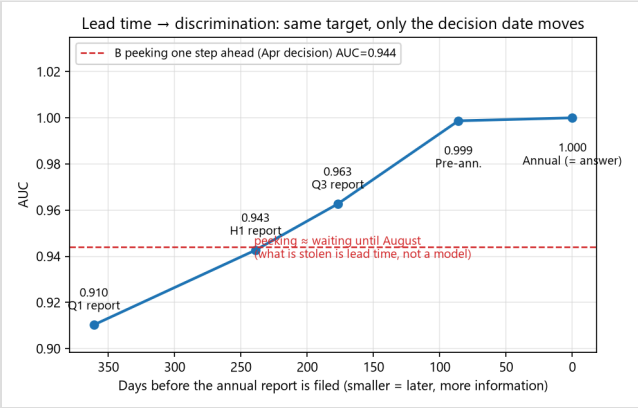
What was built

- Panel**: China A-shares 2016Q1–2026Q2 · **42 periods** · **195,707 rows** · **5,252 stocks** (four EastMoney bulk tables + the **CNINFO disclosure calendar** as the point-in-time anchor). **Task**: sample = (ticker, year) on the **Q1 report**; label = **a loss in that year's annual report**; decision time = the Q1 report's **actual filing date** (median Apr 23) ⇒ a genuine 8–12 month horizon.
- Four-arm control** (same data / model / test set; only “was the data used correctly” changes): A honest (point-in-time + walk-forward) · B peek at the next report (adds H1 figures) · C random split (training set contains future years) · D negative control (shuffled training labels); baseline = accounting persistence. Plus **five disclosure-chain steps** (Q1 → H1 → Q3 → **pre-announcement** → annual report) and a **signal timeline** of the same firms' trajectory.

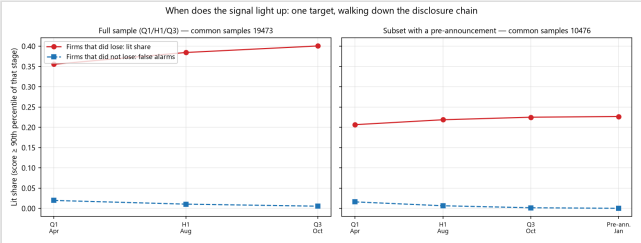
Hardest numbers

Arm	What changed	AUC	Share of the top decile that did lose
A honest	—	0.9103	84.95% (base rate 23.9% ⇒ lift 3.55)
B peek next report	features + H1 (filed 121 days later)	0.9442	92.40%
C random split	training set contains future years	0.9246	87.63%
D negative control	shuffled training labels (5 seeds)	0.4993 ± 0.0637	25.2%
Baseline: loss last year	no training	0.7512	70.4%

- The 3.4 points from “peeking”** ≈ the legitimate score of simply waiting until August (0.9442 vs **0.9427**) ⇒ **what is stolen is not a stronger model, it is lead time**: how much score would you give up to know this much earlier? (Arm A is stable: 0.891/0.896/0.922/0.928 by year; seed range 0.001.)
- When does the signal light up**: share of eventual losers already in the top decile — **Apr 35.6% → Aug 38.5% → Oct 40.1%** (false alarms 2.0% → 0.6%); **43.8% of losers never light up** ⇒ **the ceiling is not the model, it is the day the fundamentals start looking bad**.
- A trap inside the source**: EastMoney's “announcement date” is not the first filing date but the *latest* filing carrying those figures — for statements with prior-year comparatives it is **365 days late**; evidence: Moutai 2016Q1 filed 2016-04-21, EastMoney's income statement says 2017-04-25.
- Switching to the regulator's own target** (a self-built as-of “risk-warning” label: **952 events / 606 firms**): AUC **0.910 → 0.813**, but the top decile now **catches 49% of all risk-warning events** (36% for the loss label) at **lift 4.89×** ⇒ **the rarer the target, the harder the AUC** — and the more the list is worth.



Lead time → discrimination: the red line is the 3.4 “peeked” points, cutting exactly at the H1 point.



When the signal lights up: red = firms that did lose, blue = false alarms.

The one methodology worth keeping

- Never subtract two scores from different samples**: arm C's 0.9246 vs arm A's 0.9103 looks like +1.4 points, but the test sets differ; restricting A to C's samples gives **0.9122** ⇒ the real gap is **+1.24**.
- “Separable” ≠ “incremental”**: lawsuit/guarantee signals separate losers by **17.6 points** on their own, yet add only **+0.0014** AUC; and widening the list with them is **dominated by the model's own list at the same size** — that made me **retract** a verdict.
- A negative control needs a distribution** — read as macro, not pooled: a no-signal model reads **0.555** or **0.407** pooled; macro returns to **0.5**. Across stages, compare within-stage percentiles (`docs/口径.md`).

Honest boundaries

- Main table uses “annual loss”; the **as-of risk-warning label exists** as a second set of readings, but has only **20–74 positives a year** ⇒ four years must be pooled. **Delisted firms are missing; dates and panel values were both reconciled in full** (net profit matches by name only 23%, 99.6% once aligned).
- Do not read the **96.2% precision** as recall (that list covers only **40%** of losers); the pre-announcement step is a **subset**, the flash-report step is **not doable**. **Why firms get flagged is now classified** from 211 announcement PDFs: among firms flagged *without* a loss, **~64% are governance/compliance triggers** (adverse internal-control opinion, administrative penalty, frozen accounts, fund misappropriation) vs ~20% purely financial ⇒ **the trigger is not in the statements**. Caveat: only 45 such cases, keyword rules.

Entry points: `C:\dev\finlab-ml` (git · 33 commits · 10 commands, end-to-end reproducible · artefacts not committed) · **calibration** `docs/口径.md` · **findings & red lines** `docs/结论.md` · **ten documents** `docs/` · **figures** `docs/figs/` (eight)